THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS · NCERT-TYPE CHAPTER TEST SET 3

Chapter 14: **Surface Area and Volume** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Take $\pi = \frac{22}{7}$. The surface area of a sphere of radius 21 cm is: 1

(a) $6654\ \text{cm}^2$ **(b)** $5544\ \text{cm}^2$ **(c)** $6099\ \text{cm}^2$ **(d)** $4989\ \text{cm}^2$

2. Take $\pi = \frac{22}{7}$. The volume of a sphere of radius 21 cm is: 1

(a) $38808\ \text{cm}^3$ **(b)** $46571\ \text{cm}^3$ **(c)** $34927\ \text{cm}^3$ **(d)** $42689\ \text{cm}^3$

3. Take $\pi = \frac{22}{7}$. The curved surface area of a hemisphere of radius 7 cm is: 1

(a) $339\ \text{cm}^2$ **(b)** $371\ \text{cm}^2$ **(c)** $308\ \text{cm}^2$ **(d)** $277\ \text{cm}^2$

4. The unit of volume is: 1

(a) units **(b)** square units **(c)** cubic units **(d)** degrees

5. Take $\pi = \frac{22}{7}$. The curved surface area of a cylinder of radius 21 cm and height 30 cm is: 1

(a) $4357\ \text{cm}^2$ **(b)** $3563\ \text{cm}^2$ **(c)** $4754\ \text{cm}^2$ **(d)** $3960\ \text{cm}^2$

6. Take $\pi = \frac{22}{7}$. The total surface area of a cylinder of radius 10.5 cm and height 15 cm is: 1

(a) $1852\ \text{cm}^2$ **(b)** $1683\ \text{cm}^2$ **(c)** $2021\ \text{cm}^2$ **(d)** $1514\ \text{cm}^2$

7. Take $\pi = \frac{22}{7}$. The curved surface area of a cone of radius 42 cm and slant height 70 cm is: 1

(a) $9240\ \text{cm}^2$ **(b)** $11090\ \text{cm}^2$ **(c)** $8315\ \text{cm}^2$ **(d)** $10165\ \text{cm}^2$

8. Take $\pi = \frac{22}{7}$. The volume of a cone of radius 17.5 cm and height 42 cm is: 1

(a) $12127\ \text{cm}^3$ **(b)** $13475\ \text{cm}^3$ **(c)** $16172\ \text{cm}^3$ **(d)** $14823\ \text{cm}^3$

9. Take $\pi = \frac{22}{7}$. The surface area of a sphere of radius 28 cm is: 1

(a) $11829\ \text{cm}^2$ **(b)** $8870\ \text{cm}^2$ **(c)** $10842\ \text{cm}^2$ **(d)** $9856\ \text{cm}^2$

10. Take $\pi = \frac{22}{7}$. The volume of a sphere of radius 42 cm is: 1

(a) $372558\ \text{cm}^3$ **(b)** $279417\ \text{cm}^3$ **(c)** $341511\ \text{cm}^3$ **(d)** $310464\ \text{cm}^3$

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. A pyramid has a square base of side 3 cm and height 21 cm. Find its volume. 2

12. Take $\pi = \frac{22}{7}$. A cone has radius 17.5 cm and height 42 cm. Find its slant height, curved surface area and volume. 2

13. Take $\pi = \frac{22}{7}$. Find the curved surface area, total surface area of a solid hemisphere of radius 3.5 cm. 2

14. A pyramid has a square base of side 11 cm and height 15 cm. Find its volume. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Take $\pi = \frac{22}{7}$. A cone has radius 17.5 cm and height 42 cm. Find its slant height, curved surface area and volume. 3

16. Take $\pi = \frac{22}{7}$. Find the curved surface area, total surface area and volume of a solid hemisphere of radius 21 cm. 3

17. Take $\pi = \frac{22}{7}$. A cylindrical pillar has radius 7 cm and height 15 cm. Find the cost of painting its curved surface at ₹30 per 1000 cm^2. 3

OR

Take $\pi = \frac{22}{7}$. A cone has radius 7 cm and height 24 cm. Find its slant height, curved surface area and volume. 3

18. Take $\pi = \frac{22}{7}$. Find the curved surface area, total surface area and volume of a solid hemisphere of radius 10.5 cm. 3

OR

Take $\pi = \frac{22}{7}$. A cylindrical pillar has radius 21 cm and height 20 cm. Find the cost of painting its curved surface at ₹20 per 1000 cm^2. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. A rectangular sheet 132 cm long and 5 cm wide is rolled so that its length forms the circumference of the base of a cylinder of height 5 cm. Find the radius, the curved surface area and the volume (Take $\pi = \frac{22}{7}$). 5

OR

A solid metal sphere of radius 25 cm is melted and recast into small solid spheres of radius 5 cm. Find the number of small spheres and compare the total surface area of the small spheres with the surface area of the big sphere. 5

20. A rectangular sheet 88 cm long and 16 cm wide is rolled so that its length forms the circumference of the base of a cylinder of height 16 cm. Find the radius, the curved surface area and the volume (Take $\pi = \frac{22}{7}$). 5

OR

A solid metal sphere of radius 30 cm is melted and recast into small solid spheres of radius 5 cm. Find the number of small spheres and compare the total surface area of the small spheres with the surface area of the big sphere. 5

END OF QUESTION PAPER — **Total: 40 Marks**